STUMPRIDER · DESIGN DECISIONS REGISTER

StumpRider design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25 ("i approve of your recommendations"): O6 (A), O7 (A), O8 (A) and O9 (A) moved to Decisions made (SMR-DDR-006); O10 (C) approved but stopped while carrying it through and returned to Amish; bamboo local variant USD 66.90
0.4	2026-10-24	Amish Chadha	Kit 1.9.1 upgrade: F1 (hook head J of 12 mm bar) and F2 (option C tines) decided under Amish's standing instruction of 2026-10-23 after the STEM check and its independent review; O6 to O10 proposed, awaiting Amish; detailed CAD of every option recorded
0.3	2026-10-04	Amish Chadha	O5 decided by Amish (round-3 decision 2A, SMR-DDR-004) and moved to Decisions made; no open decisions
0.2	2026-10-03	Amish Chadha	O1 to O4 decided by Amish (10B, 11A then C, 12A, 13C; SMR-DDR-003); one new open decision on the aluminium prototype's carried mass; items to

			confirm and value engineering updated
0.1	2026-10-03	Amish Chadha	Register opened; design decisions made under Amish's 2026-10-03 pre-approvals; four requirement decisions proposed, awaiting Amish

StumpRider design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in `docs/REVIEW.md`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list decisions.

> **Safety:** Several entries set safety limits (the pin rating and its calibration, kneeling, slackening the line, the interim rule on small canoes, life jackets). Each takes the conservative option and names the evidence that would relax it. StumpRider is never a reason to keep children on boats and never replaces a life jacket.

Open decisions

O1 to O4 were decided on 2026-10-03, O5 (R7 on the aluminium prototype) on 2026-10-04, and O6 to O9 on 2026-10-25; all are under Decisions made. O10 is open again: Amish approved its recommendation (C) on 2026-10-25, but carrying it through showed that the gusset does not do what the recommendation said without breaking PR-007 (see the row and `docs/REVIEW.md`).

Table 1. Open decisions (proposed, awaiting Amish).

#	Decision	Options	Recommendation	Record
O10	Hook head with the bight caught on the tip leg (SMR-STM-001 S3b): factor 1.58 on yield at the bend, under the 2 x rule (PR-012); no yield before the pin breaks. **Approved as recommended (C, a small gusset outside the bend) by Amish on 2026-10-25; stopped while carrying it through:** the tip-leg load puts the same moment (load x 60 mm) into the straight shank between the bend and the flat, factor 1.87 [S13a], which a gusset at the bend does not reach; a gusset to the flat (about 0.06 to 0.10 kg, estimate) takes the bamboo variant past 4.0 kg (PR-007, 0.05 kg margin)	A: accept, with the bottom of the J as the design case and the tip-leg case recorded (no cost). B: 14 mm bar (bend factor about 2.3, shank about 3.0; about 0.09 kg more, PR-007 fails). C: a gusset at the bend (about 0.02 kg; shank stays 1.87). D (new): a gusset from the bend to the flat (shank and bend over 2; about 0.06 to 0.10 kg, PR-007 fails)	**None now.** The approved fix does not reach the 2 x rule on paper; the choice is between accepting the tip-leg case (A) and giving up PR-007's margin (B or D), a trade-off for Amish	SMR-STM-001 S3b; SMR-CAL-001 [S13, S13a]

To confirm when parts are bought

These are facts that can only be settled with real parts, a real canoe or the first partner. None changes a decision; each may change a size or a limit.

Table 2. Items to confirm.

#	What to confirm	Why it matters	Source
1	Shear strength of the chosen wire reel (48 MPa assumed): ten pins broken on CalRig	Pin rating and its band (R3)	SMR-CAL-001, A1; SMR-DDR-002, A2
2	38 mm tube with a 32.4 mm bore, or how much opening 38 x 3 tube needs	Sleeve fit	SMR-DDR-002, A1
3	Size, mass and crew of the partner's canoes; gunwale thickness and clear side depth	Heel check (R3) and crutch fit	SMR-CAL-001, A2 to A5; SMR-DDR-002, A3 and A4
4	How often nets hook and how often they wrap, from partner crews; how hard the hook head must pull to work a wrap back half a turn	R1 and the hook head	SMR-PRB-001; SMR-DDR-003
5	Friction of the partner's netting on wet bark (0.3 to 0.5 assumed)	Freeing pull (R1)	SMR-CAL-001, A7
6	Local prices for aluminium tube, bamboo, steel tube, galvanising and foam near the first fishery	R10 on the bamboo local variant	`bom/bom.csv`, `bom/bom-bamboo-variant.csv`
7	Bent ring holds 112 mm mean diameter within 2 mm	Joint gaps and ear alignment	SMR-DDR-002, A5
8	Mesh damage at the pin rating on 90 to 150 mm mesh	R4	SMR-REQ-001
9	Stiffness of local bamboo culms: each culm passes the bend test (5 kg at the middle of a 1.4 m span sags 2.9 mm or less); culm diameters 34 to 38 mm slide in the 37 mm ferrule	Bamboo pole buckling (factor 1.49 at E 15 GPa, 1.0 at 10 GPa)	SMR-CAL-001 [V5], [V6], [V10]; SMR-DDR-003
10	Borax and boric acid treatment and zinc-painted ferrules hold up for 6 months wet	R8 on the bamboo local variant	SMR-DDR-003
11	Canoe size classes at the partner's landings and a measured pin for each class (heel tests at TRL 4)	R3 for canoes under 7 m	SMR-DDR-003
12	Workshop time for one kit with the hook head (8.1 h estimated against one day)	R9	SMR-CAL-001 [T1]

Value engineering

Value-engineering target: USD 2,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 299.50 for the prototype run of three aluminium kits, one bamboo pole set and calibration wire (USD 1,700.50 under the target). One aluminium prototype kit is USD 95.50 with the hook head; one bamboo local variant kit with its own crutch and the jigging weight shared by five canoes is USD 66.90 (USD 54.90 when the crutch was shared too; Amish's decision O8 of 2026-10-25). Main cost drivers and savings worth trying:

- In the aluminium kit the largest lines are the pole, sleeves and rivets (30 %), galvanising (13 %) and the crutch (13 %); the bamboo local variant removes the first; since O8 each canoe keeps its crutch, so only the jigging weight is shared.
- Savings still worth trying on the local variant: galvanising a batch of kits together, which spreads the galvaniser's minimum lot; zinc-rich paint only where no galvaniser is near (about USD 9, at some risk to R8); local prices for bamboo and steel tube.

- Smith's labour (about 8.1 hours a kit) is not in the parts cost; a partner workshop making kits in batches would cut it.

Decisions made under the standing instruction

Amish's standing instruction of 2026-10-23: when a STEM check row fails, propose and implement the best fix for the design and its constructability in the same session and record it here; a fix that changes the working principle, pitch or scope goes to Amish instead. Amish can reverse any of these.

Table 3. Fixes under the standing instruction.

#	Date	What failed	Options weighed	Fix and cost	Record
F2	2026-10-24	Option C (concept only): grapnel tines of 12 mm bar at 1.91 times the link's top load, under the 2 x rule (independent review)	(a) 14 mm tines; (b) shorter tines (less bite)	(a): 14 mm tines in the option C model (`cad/src/model.py` gr_tine_d); factor 3.0. No BOM, drawing or build plan exists for option C	SMR-STM-001 C2; decided under Amish's standing instruction of 2026-10-23
F1	2026-10-24	Hook head J of 10 mm bar: 1.86 times as strong as the top of the pin band with the bight in the J (curved-bar bending), under the rule that every part but the pin is at least twice as strong (PR-012)	(a) 12 mm bar, the ring's stock; (b) a smaller bend radius (narrows the throat below a doubled 16 mm line with room); (c) a welded gusset (more welding, a snag point)	(a): J of 12 mm bar, same 30 mm bend radius (48 mm throat), butt welded to the flat's lower edge with a fillet all round (a 12 mm bar cannot lap centrally on a 6 mm flat). Factor 3.1; no cost change (ring offcuts); hook head 0.21 to 0.26 kg, bamboo variant carried 3.95 kg (PR-007 still met). Model PARAMS hook_bar_d 10 to 12; BOM line 21; SMR-DWG-001 P4 and SMR-DWG-111; SMR-BLD-001 3.11; SMR-CAL-001 [S11 to S14]	SMR-STM-001 S3; decided under Amish's standing instruction of 2026-10-23

Record of the kit 1.9.1 upgrade (2026-10-24)

Under Amish's instructions of 2026-10-24 ("I want detailed TRL3 CAD", "Detail every option", "Kit 1.9.1 rollout - proceed to rollout across the other repos"): `cad/src/model.py` gained the kit 1.9 interface, option A is modelled at D3 with every bolt, nut, washer, clevis pin, R-clip, rivet and weld bead, and options B and C are modelled at D3 in `cad/src/smr_options.py`. No design dimension of option A changed except F1. The BOM's item names now start with their line number, because `.kit/cad_check.py` reads line numbers from the item column (kit bug noted in `docs/REVIEW.md`); quantities, specifications and prices are unchanged except line 21 (F1).

Decisions made

Table 4. Decisions made.

Date	Decision	Decided by	Record
2026-10-03	TRL 2 review items D1 to D10: hinged two-half ring with a pinned gate; three-section aluminium pole for the prototype; one shear pin rating, 2.0 mm soft aluminium, 302 N nominal, calibrated per reel; the pin carries push and pull; operating rules (kneel, slacken, life jackets, tether tied off, cut rather than capsize, interim rule on canoes under 7 m); jigging weight pinned to the tang; foam float winder; clamp-on roller crutch; handed out only through child-protection partners, first candidates to approach (Challenging Heights, a Volta landing-site committee through the Fisheries Commission, a Ghanaian university fisheries department; none approached); requirement targets kept	Amish, pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." and "Proceed with the remaining 15 scaffolds"	SMR-DDR-001
2026-10-03	Design for construction, changes C1 to C12, and assumptions A1 to A5	Amish, same pre-approvals	SMR-DDR-002
2026-10-03	Safety stops S1 to S5 and the first checks, including pin calibration and a moored heel test before any use on a snag (conservative; a gate, not relaxed)	Amish, same pre-approvals	SMR-BLD-001, sections 5 and 6
2026-10-03	O1 (R1, 10B): add a hook head that pins into the same fork (about USD 3), with the unwinding technique taught. O2 (R3, 11A then C): keep the rule (canoes 7 m or longer, three crew) and plan heel tests by canoe size with a pin per class at TRL 4. O3 (R7, 12A): the crutch stays clamped on the canoe and the jigging weight at the landing (4.02 kg carried as proposed; 4.28 kg as built with the hook head and corrected masses). O4 (R10, 13C): a bamboo-pole local variant with a shared crutch and weight (USD 52.90 as proposed; USD 54.90 with the hook head) beside the aluminium prototype	Amish, 2026-10-03: "i agree with all the 46 recommendations you provided. please proceed."	SMR-DDR-003
2026-10-03	Appearance model departures for the renders: a plank deck and a forearm and hand beside the packed kit; a short net line over a branch and a cut-short pole for the detail view; the shear pin shown red	Amish, same pre-approvals	docs/REVIEW.md, TRL 3 section
2026-10-04	O5, R7 (2A): keep the 32 x 2 aluminium pole; R7 is judged on the bamboo local variant (3.89 kg, met); the aluminium prototype's 4.28 kg is recorded, not a failure of R7; the prototype is weighed at TRL 4	Amish: "For round 3, I agree with all your proposed recommendations"	SMR-DDR-004
2026-10-25	O6 (A): keep the 2.0 mm pin; 5 deg is the static heel limit; "lift steadily, never jerk" is added to the operating rules (stop S4a) and the tag; the TRL 4 moored heel test measures the roll period and the peak heel under a quick pull; a 1.7 mm pin only if that test shows a real exceedance. No cost	Amish, 2026-10-25 12:43 CT: "i approve of your recommendations"	SMR-DDR-006, item 1
2026-10-25	O7 (A): concept selection, option A (rider ring on a shear-pin pole, with B's jigging already in it);	Amish, as above	SMR-DDR-005;

Date	Decision	Decided by	Record
	`concept_chosen: A`; SMR-DDR-005 accepted. The rule "an option scoring 1 on a High UR is not eligible", offered beside the options, was not part of the recommendation and is not adopted		SMR-DDR-006, item 2
2026-10-25	O8 (A): each canoe keeps its own crutch; only the jigging weight is shared by five canoes. PR-007 unchanged (3.95 kg); bamboo local variant USD 66.90 a kit (estimate; USD 54.90 before)	Amish, as above	SMR-DDR-006, item 3; SMR-CAL-001 [V7, V8]
2026-10-25	O9 (A): keep the 7 m, three-crew rule as a screen and add a measured criterion at the TRL 4 heel test: displacement x GM of at least about 260 kg m, with a margin set from that test; PR-003 restated	Amish, as above	SMR-DDR-006, item 4; SMR-CAL-001 s3